$3x^2+2xy+3y^2+2x-6y+\frac{25}{2}=0$, write the equation in standard form

We know, the generalized conic equation is

$ax^2+2hxy+by^2+2fg+2fx+c=0$

$\therefore\ a=3,\ h=1,\ b=3,\ g=1,\ f=3,\ c=\frac{25}{2}$

To remove the xy term we need to turn the axes by angle θ, where

$$\tan 2\theta = \frac{2h}{a-b} = \frac{2(1)}{3-3} = \infty$$

$$\Rightarrow 2\theta = \tan\infty = \frac{\pi}{2}$$

$$\therefore\ \theta = \frac{\pi}{4}.$$

To rotate the ellipse the equation we need to use would be

$x = x'\cos\theta - y'\sin\theta$

$y = x'\sin\theta + y'\cos\theta$

$\theta = \pi/4$

$x = x'\cos\pi/4 - y'\sin\pi/4$

$\boxed{\therefore x = \frac{x'}{\sqrt{2}} - \frac{y'}{\sqrt{2}}}$

$y = x'\sin\pi/4 + y'\cos\pi/4$

$\boxed{\therefore y = \frac{x'}{\sqrt{2}} - \frac{y'}{\sqrt{2}}}$

Substituting the values of x & y in the ellipse equation.

$$3\left(\frac{x'}{\sqrt{2}}-\frac{y'}{\sqrt{2}}\right)^2+2\left(\frac{x'}{\sqrt{2}}-\frac{y'}{\sqrt{2}}\right)\left(\frac{x'}{\sqrt{2}}+\frac{y'}{\sqrt{2}}\right)+3\left(\frac{x'}{\sqrt{2}}+\frac{y'}{\sqrt{2}}\right)^2$$
$$+2\left(\frac{x'}{\sqrt{2}}-\frac{y'}{\sqrt{2}}\right)-6\left(\frac{x'}{\sqrt{2}}+\frac{y'}{\sqrt{2}}\right)+\frac{25}{2}=0$$

Suppressing the primes from the above equation

$$3\left(\frac{x^2}{2}-2\cdot\frac{x}{\sqrt{2}}\cdot\frac{y}{\sqrt{2}}+\frac{y^2}{2}\right)+2\left(\frac{x^2}{2}-\frac{y^2}{2}\right)+3\left(\frac{x^2}{2}+2\cdot\frac{x}{\sqrt{2}}\frac{y}{\sqrt{2}}+\frac{y^2}{2}\right)$$
$$+\frac{2x}{\sqrt{2}}-\frac{2y}{\sqrt{2}}-\frac{6x'}{\sqrt{2}}-\frac{6y'}{\sqrt{2}}+\frac{25}{2}=0$$

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$$\Rightarrow 3\left(\frac{x^2}{2} - \frac{2xy}{2} + \frac{y^2}{2}\right) + 2\left(\frac{x^2}{2} - \frac{y^2}{2}\right) + 3\left(\frac{x^2}{2} + \frac{2xy}{2} + \frac{y^2}{2}\right)$$

$$+ 2\left(\frac{x}{\sqrt{2}} - \frac{y}{\sqrt{2}}\right) - 6\left(\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}\right) + \frac{25}{2} = 0$$

$$\Rightarrow \frac{3x^2}{2} - \cancel{3xy} + \frac{3y^2}{2} + x^2 - y^2 + \frac{3x^2}{2} + \cancel{3xy} + \frac{3y^2}{2}$$

$$+ \frac{2x}{\sqrt{2}} - \frac{2y}{\sqrt{2}} - \frac{6x}{\sqrt{2}} - \frac{6y}{\sqrt{2}} + \frac{25}{2} = 0$$

$$\Rightarrow 4x^2 + 2y^2 - \frac{4x}{\sqrt{2}} - \frac{8y}{\sqrt{2}} + \frac{25}{2} = 0$$

$$\Rightarrow 4\left(x^2 - \frac{x}{\sqrt{2}}\right) + 2\left(y^2 - \frac{4y}{\sqrt{2}}\right) + \frac{25}{2} = 0$$

$$\Rightarrow 4\left(x^2 - 2.\frac{x}{2\sqrt{2}} + \frac{1}{(2\sqrt{2})^2} - \frac{1}{(2\sqrt{2})^2}\right)$$

$$+ 2\left(y^2 - 2.\frac{2y}{\sqrt{2}} + \left(\frac{2}{\sqrt{2}}\right)^2 - \left(\frac{2}{\sqrt{2}}\right)^2\right) + \frac{25}{2} = 0$$

$$\Rightarrow 4\left(x - \frac{1}{2\sqrt{2}}\right)^2 - \frac{1}{2} + 2\left(y - \frac{2}{\sqrt{2}}\right)^2 - \frac{2\times 4}{2} + \frac{25}{2} = 0$$

$$\Rightarrow 4\left(x - \frac{1}{2\sqrt{2}}\right)^2 + 2\left(y - \frac{2}{\sqrt{2}}\right)^2 = -17.$$

$$\frac{\left(x - \frac{1}{2\sqrt{2}}\right)}{\frac{-17}{4}} + \frac{\left(y - \frac{2}{\sqrt{2}}\right)^2}{\frac{-17}{2}} = 1$$

[This should be negative]

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